

Interactive Digital Art and Design SE608

Project Proposal Year IV

Student Name:	Caoimhe O'Brien	Student Number:	C00297178
Working Title:	"The POUR Project"		
Description:	Synopsis: "The POUR Project" is a multi-level first-person RPG game where the player will defeat different obstacles that various user bases face while navigating the online world. It will show why digital accessibility equality is important. One of the main ways Accessibility is enforced is through WCAG 2 (Web Content Accessibility Guidelines). The Accessibility Issue this Game Will Address: Within these guidelines there are 4 principles: Perceivable, Operable, Understandable and Robust which is where "POUR" comes from. For my project, I will make at least 4 levels, with each level containing a boss who will represent one of the 4 "POUR" principles. What the Game Will Look Like: The "Perceivable" principle of WCAG dictates that digital content must be built so that users can easily see or hear it, regardless of their visual or auditory impairments. Some ways this principle is implemented into web content is through requiring high colour contrast, alt text for images and correct tagging for screen-readers. To highlight the obstacles in this pillar, I will make a level where the player has to combat the Level Boss whose power is to lower the colour contrast in the game so it's hard to make out where to attack/ where the enemy is. The player will also rely on audio cues on this level but the minions (smaller enemies for each of the principles) can interrupt the audio cues. This will highlight what's it's like for people with low vision to access websites with improper colour contrast and also what it's like to use a screen reader, especially when a page is improperly tagged. The "Operable" principle of WCAG requires interfaces to be designed in a way that everyone can easily move through them, control them and press buttons without any barriers. Some ways this principle is implemented into web content is through allowing keyboard navigation, creating intuitive/ easy-to-use layouts and allowing users plenty of time to complete tasks. The player has to combat the Level Boss whose main power is making the keys not do what they're meant to do. So, when the Boss attacks the player, it might move it's "up" function to a different key (instead of it being the "W" key, it might be moved to the "K" key) and then the player would have to figure out what key they need to use now and this would all be under a time limit. This would highlight what it is like when websites don't follow proper control conventions and when users are required to complete tasks under a time limit. The "Understandable" principle of WCAG means that websites and apps must		

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be clear and predictable so that users don't get confused when completing a task. Some ways this principle is enforced is through using simple language, having navigation behave as expected and providing friendly/useful/specific error messages for users. For this level, the player will have to complete riddles that the Boss gives them. However, the Boss doesn't give the riddles in plain English and/or mixes up words. The answers to the riddles will be simple to guess but the real task of the level is figuring out what the Boss is actually asking you. Once they know the answer to the riddle, they have to try and give the answer to the Boss but the answer submission won't work as expected so it will be another riddle on top of the first riddle. This would highlight what it's like for users to experience what it's like to navigate a website when the website does not use plain, simple English and also does not abide by predictable navigation behaviours.

The "Robust" principle of WCAG means that the underlying code works flawlessly across all types of technology, and that it can adapt to other technologies that may be created in the future. Some ways that this principle is enforced is through creating responsive designs so that the content can adapt to any screen size, it can run on a multitude of systems (such as different web browsers) and that it works with assistive technologies (such as working with a screen reader without glitching). In this level, the user will combat with the level Boss who can change the player's view and glitch out certain combat moves (as if it were an assistive technology or a different browser). This would highlight what it's like to use a website that is not responsively designed or designed with only 1 output in mind (i.e. only for desktop or only for Chrome).

The game ends when you defeat all 4 Bosses, bringing Accessibility to the WWW. There will also be an informational page that tells you about WCAG (similarly to how I've done here) and links to useful resources.

Technical Elements:

To create the game, I want use Godot as the game engine rather than Unity or UPBGE because Godot is optimised for multitude of devices such as mobile, Linux machines, desktop, etc. as well as having a built-in accessibility infrastructure (AccessKit) which is in keeping with the theme of accessibility. It is also optimised for both 2D and 3D art which I am open to exploring. It allows you to code your game in C++(or C# rather than using its own scripting language (GDScript)) and I have coded games in C++ before. Godot allows for 2D skeletal animation (rigging) for the actual gameplay and cutscenes. Godot is also I will also be hosting this game online through itch.io and have a downloadable package through my portfolio (for windows, linux, mac, apk (apk) and .zip (for all others)).

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	Art Elements: To create the assets for this game, I will use a mixture of Blender (for the environment/backgrounds, and Krita (for the 2D sprites). I also aim to create informational cutscenes using Krita/Blender to further inform the user on the importance of digital accessibility. Design Elements: For this game, I will be designing various menus and interfaces for the interactive experiences as well as creating a section on my portfolio website to allow people to play and download the experience onto their own machines. User Testing: For my user testing, I will ask a group of participants to play through a level giving verbal feedback. At the completion of the level, I will ask them to complete a short survey to investigate whether the key theme of the game (accessibility) was shown and if they felt impacted by the issues highlighted in the level. I will not keep any identifying data on any of the user testers and the survey will be anonymous.
Reasons for selecting project:	This project will build on the work I did as part of my work placement in regard to accessibility. It is an important topic in the industry at the moment and this game aims to bridge the gap in understanding as a lot of the general public take accessibility for granted and/or are unaware of the impact of inaccessible web content.
Proposed research content:	https://dl.acm.org/doi/10.1145/3802826
External links (if applicable):	https://www.ohchr.org/en/instruments-mechanisms/instruments/convention-rights-persons-disabilities https://www.w3.org/TR/WCAG22/ https://www.iso.org/standard/51341.html https://www.iso.org/standard/71953.html https://nda.ie/publications/a-user-centred-exploration-of-the-assistive-potential-of-digital-voice-assistants-and-internet-of-things Vision for the Godot Engine – Godot Foundation Godot 4.5 improves accessibility support, including screen readers - Can I Play That?
Hardware requirements:	Lab computer, drawing tablet (already owned)
Software requirements:	Godot, Blender, C#, HTML/CSS/JS, GitHub, OneDrive, Krita, Itch.io
Other requirements:	User testing (Ethics form)
Actions Taken	Highlighted the accessibility issue that the game will be highlighting/addressing Highlighted why Godot is the most suitable game engine and why I want to make this a “game”



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	Proposed why a “game” format is the suitable medium for this project
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Signed:

Date: 25/09/2026

Coimhe O'Brien

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Approved/Not Approved:	
Reasons for not approving project:	
Conditions attached to approving project:	
Approved/Not Approved:	
Name of Supervisor:	
Signed: _____	Date: _____